Assignment 1: Regional Projections and the Nepal Flood

Due: October 27, 2026 · Submission: two comments on the course GitHub Discussions

Overview

This assignment has two parts, and each is submitted as a comment in its own GitHub Discussion thread.

- Part 1: you read climate model projections for a region set by your roll number and compare them with your classmates' readings.
- Part 2: you add new evidence to our ongoing discussion of the 2026 Bhote Koshi flood in Nepal.

If you prefer a username that does not show your name, inform the TAs of your username before the deadline.

AI use. You may use AI tools, as elsewhere in this course. If you do, add one line at the end of each comment saying which tool you used and for what purpose. The numbers you report must come from your own work in the Atlas. The comments you respond to must be ones you have read in the threads.

Part 1: Regional projections from the IPCC Interactive Atlas (10 marks)

Thread: https://github.com/LUMS-WIT/climate-futures/discussions/6
Tool: https://interactive-atlas.ipcc.ch/regional-information

Start by reading the opening post in the thread. It summarises what AR6 concluded from climate models, and your synthesis will be tested against it.

Use the CMIP6 dataset throughout. The last two digits of your roll number decide your region and your future.

Last digit: your region (AR6 reference region)

Digit	Region	Digit	Region
0	South Asia (SAS)	5	West Africa (WAF)
1	Tibetan Plateau (TIB)	6	East Southern Africa (ESAF)
2	West Central Asia (WCA)	7	North Eastern South America (NES)
3	Arabian Peninsula (ARP)	8	Central North America (CNA)
4	Mediterranean (MED)	9	Northern Europe (NEU)

Second last digit: your future

Digit	Setting
0 or 1	SSP1-2.6, medium term (2041–2060)
2 or 3	SSP2-4.5, long term (2081–2100)
4 or 5	Global warming level of 2°C
6 or 7	Global warming level of 3°C
8 or 9	SSP5-8.5, long term (2081–2100)

Example. Roll number 26100147 has a last digit of 7 and a second last digit of 4. That student uses North Eastern South America at a global warming level of 2°C.

For every reading below, report three things: the median, the 10th to 90th percentile range, and the baseline period you selected.

Tasks

1. **Temperature.** Report the annual mean temperature change for your region. Then switch the region to South Asia, keeping every other setting the same. Report the ratio of your region's warming to South Asia's.
2. **Precipitation.** Report the annual total precipitation change in percent. State whether the Atlas marks the signal as robust, and explain what the hatching tells you about model agreement.
3. **Heavy rainfall.** Report the change in maximum one-day precipitation (Rx1day). Using the model spread view, count how many models project an increase and how many project a decrease.
4. **Your choice.** Choose one further index that matters most for your region, for example TX35, consecutive dry days or frost days. Explain your choice in two sentences and report its change.
5. **Synthesis (200 to 300 words).** Explain where your results agree with or depart from the AR6 findings in the opening post. Finish with what your results imply for how a government in your region should weigh projected temperature against projected precipitation when it plans.

Format. Begin your comment with one line giving your roll number, region and setting. Then answer tasks 1 to 5 in order. Attach one screenshot that shows the Atlas settings panel for your region; you can drag the image straight into the comment box.

Part 2: Evidence on the Nepal flood (10 marks)

Thread: https://github.com/LUMS-WIT/climate-futures/discussions/5

Read the opening post and every comment already posted before you write. Your comment should bring in evidence the thread does not yet contain. It could be a historical event, a comparable event in another mountain region, a physical mechanism, or an angle on exposure, early warning or governance.

Your comment (under 300 words) must:

- Link to the specific earlier comment you are building on or disagreeing with.
- Cite at least one source whose link does not already appear anywhere in the thread.
- State in one sentence what your case has in common with the Bhote Koshi event, and in one sentence what is different.
- Summarise in two or three sentences what the thread as a whole now suggests about the cause of the flood, and what remains uncertain.
- End with one question the thread has not yet asked.

Post a new top-level comment rather than a reply, so that each contribution stands on its own. Comments that repeat a source, event or argument already in the thread will not earn marks for novelty. Posting order is visible from the time stamps, so an earlier comment takes precedence.

Marking criteria

Each part is marked out of 10.

Criterion	Part 1	Part 2
Accuracy and completeness of Atlas readings (Part 1), or novelty and quality of evidence (Part 2)	4	4
Quality of synthesis and engagement with classmates' posts	3	3
Clarity of reasoning about uncertainty	2	2
Closing implication (Part 1) or closing question (Part 2)	1	1

Late comments follow the course late policy. Edits made after the deadline will not be considered.